

The Politics of Anti-Corruption: Bureaucratic Sabotage and Collusion

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DRAFT FOR MPSA: Comments Welcome

Abstract

Anti-corruption reforms often strengthen political enforcement and improve monitoring, yet reform-minded governments do not always preside over less corruption. I argue that the missing link is the behavior of unelected intermediaries who supervise subordinates and influence whether corruption is restrained or tolerated. I develop a dynamic model in which a rent-seeking bureaucratic intermediary chooses the level of corruption before an election, knowing that current corruption affects both current rents and the probability that the incumbent remains in office. A lenient incumbent encourages *strategic collusion*: the intermediary restrains corruption to preserve a favorable principal. A tough incumbent incentivizes *strategic sabotage*: the intermediary permits more corruption to increase the likelihood of turnover. These opposing incentives imply a non-monotonic relationship between enforcement and corruption. Starting from low enforcement, tougher enforcement can increase corruption; beyond a threshold, the direct deterrent effect dominates and corruption falls. The model also shows that better information and higher state capacity need not reduce corruption or improve voter welfare monotonically. More broadly, the paper shows how unelected officials can subvert electoral accountability by shaping the underlying level of corruption itself.

Introduction

The modern state fights corruption through politics as much as through law. Voters elect leaders who promise to clean up the bureaucracy, strengthen oversight, and raise the cost of graft. Standard principal-agent accounts give that promise a clear logic: tougher principals and better accountability should discipline self-interested agents and reduce corruption (Shleifer and Vishny, 1993; Rose-Ackerman, 2013; Groenendijk, 1997; Barro, 1973; Fearon, 1999; Williams, 2021). Yet the political record is harder to square with that monotone view. Reform-minded politicians often come to office on anti-corruption platforms only to govern through periods marked by continuing graft inside ministries, state-owned firms, procurement offices, and prosecutorial agencies. By contrast, politicians who are permissive, compromised, or later revealed to be corrupt can preside over bureaucracies that appear

comparatively orderly. If stronger political will simply raised the expected cost of misconduct, this pattern should not arise.

This paper argues that the missing link is the behavior of unelected intermediaries: agency heads, senior prosecutors, procurement directors, inspectors, and managers who supervise subordinates and shape whether corruption is deterred or tolerated. These actors often do not bear the electoral cost when corruption becomes politically salient. They do, however, control whether lower-level corruption is contained or allowed to spread. That creates a political margin absent from standard models. A tough elected principal threatens future rents, influence, or protection. A permissive principal preserves them. Intermediaries may, therefore, respond to a hostile principal not by reducing corruption, but by allowing more of it below them. The same intermediaries may discipline subordinates more aggressively under a favorable principal, precisely because limiting corruption helps shield an incumbent from whom they continue to benefit.

Concrete episodes fit both sides of this logic. South Africa under Zuma and Ramaphosa illustrates the sabotage side. The Zondo inquiry documented systemic graft during Zuma’s presidency, yet Ramaphosa’s reformist turn did not produce an immediate bureaucratic clean-up; instead, it was met with persistent resistance from entrenched networks and continuing corruption inside the state (PARI, 2022; South African Institute of International Affairs, 2024). Nigeria under Buhari is similar. Buhari was elected largely on an anti-corruption pledge, but his presidency soon confronted repeated allegations involving senior officials and administrative agencies, while broader enforcement remained weak (Hoffmann and Patel, 2017; Reuters, 2018; U4 Anti-Corruption Resource Centre, 2023). The collusion side appears in more centralized patronal systems. In Russia and Serbia, corruption at the top has coexisted with strong incentives for loyal elites to keep lower-level predation politically contained, because visible bureaucratic disorder threatens the principals who supply contracts, protection, and continued rents (Lanskoy and Myles-Primakoff, 2018; Dawisha, 2015; Bertelsmann Stiftung, 2024; Freedom House, 2024). The point is not that permissive principals eliminate corruption. It is that they can induce intermediaries to manage it, whereas hostile principals can induce intermediaries to tolerate more of it.

The question is difficult for three reasons. First, more corruption under a reformer need not reflect strategic resistance; it may instead reflect inherited networks, administrative transition, or simple mean reversion. Second, observed corruption may rise because monitoring improves, even if underlying corruption does not. A theory centered only on exposure therefore misses an important distinction. Third, the relevant actors are neither ultimate political principals nor ordinary street-level bureaucrats. They are supervisory intermediaries who can influence subordinates’ behavior while remaining partially insulated from electoral sanction. A persuasive account must therefore explain why such actors would ever permit more corruption under some principals than under others.

I formalize this idea in a dynamic model of electoral selection. For tractability, the intermediary is a single Bureaucrat. Substantively, he can be interpreted as an agency head or senior administrator who influences whether subordinates engage in corruption and how aggressively such behavior is contained. Before an election, the Bureaucrat “chooses” corruption, knowing that current corruption affects both current rents and the identity of the politician he expects to face tomorrow. Voters, who want to limit corruption, do not observe corruption directly. Instead they see a noisy public signal and decide whether to retain the

incumbent. Corruption therefore serves two purposes at once: it generates immediate rents, and it helps determine future political control.

The model generates two distortions. When the incumbent is lenient, the Bureaucrat engages in *strategic collusion*: he restrains corruption in order to preserve a favorable principal. When the incumbent is tough, he engages in *strategic sabotage*: he allows more corruption in order to induce turnover. These opposing incentives imply that the relationship between enforcement and corruption is non-monotonic. Starting from low levels of enforcement, marginal tightening can increase equilibrium corruption because it weakens the value of retaining the incumbent and strengthens the incentive to engineer replacement. Only beyond a threshold does the direct cost of graft dominate and corruption fall.

The same logic extends to information and state capacity. Better information does not merely help voters discipline corruption; it also makes realized corruption a more effective instrument for influencing electoral survival. As a result, greater information quality can increase not just the visibility of wrongdoing, but the equilibrium level of true corruption. State capacity is likewise not uniformly disciplining. Higher capacity directly raises the cost of graft, but it also changes the intermediary's political incentives. When capacity is initially low, improving it can weaken the intermediary's incentive to restrain corruption under favorable principals before the direct deterrent effect becomes dominant. Over that range, stronger state capacity can raise equilibrium corruption rather than reduce it. Once electoral accountability is endogenized, these same forces imply that neither better information nor higher state capacity need improve voter welfare monotonically.

The rest of the paper proceeds as follows. The next section situates the argument within the literatures on corruption, political agency, and sabotage. I then present the model and characterize the equilibrium distortions it generates. The following sections study electoral accountability, comparative statics, and welfare. The final section discusses implications for anti-corruption reform and organizational design.

Related Literature

This paper connects two literatures: work on corruption and political agency, and recent work on strategic policy sabotage.

A large literature studies corruption as a principal-agent problem in which officeholders or bureaucrats trade off illicit rents against the probability and cost of detection. [Shleifer and Vishny \(1993\)](#) provides the canonical formalization of how organizational structure and market competition shape bribe extraction; see also [Rose-Ackerman \(2013\)](#); [Groenendijk \(1997\)](#). In these models, stronger monitoring or harsher sanctions typically reduce graft monotonically. Related work has shown, however, that this logic can break down once enforcement interacts with collusion and political incentives. In particular, [Bueno de Mesquita and Hafer \(2008\)](#) show that when government and mafia can collude, harsher sanctions against the mafia can make the government less willing to challenge it. Political agency models likewise emphasize how voters use noisy performance signals to select competent or honest incumbents ([Barro, 1973](#); [Fearon, 1999](#); [Williams, 2021](#)), while a growing literature studies how state capacity and institutional design shape enforcement ([Bersch, Praça and Taylor, 2017](#)). My paper departs from these frameworks by focusing on unelected intermediaries inside the

state who can permit or restrain subordinate corruption in order to influence which elected principal they will face tomorrow. This endogenizes the relationship between enforcement, corruption, and electoral selection.

The paper is also closely related to recent theoretical work on sabotage. [Gieczewski and Li \(2022\)](#) and [Hirsch and Kastellec \(2022\)](#) study sabotage by opposition actors and show how its timing and intensity depend on electoral incentives and incumbent popularity. Related formal work analyzes how organized actors can degrade implementation, withhold effort, or otherwise worsen outcomes to affect political or legislative payoffs ([Patty, 2016](#); [Fong and Krehbiel, 2018](#); [Delgado-Vega and Shaver, 2024](#)). Most closely, [Heo and Wirsching \(2024\)](#) develop a theory of bureaucratic sabotage in which anti-reform bureaucrats lower service quality in order to influence voters.

My paper differs from this work in both substance and mechanism. Substantively, the policy at issue – anti-corruption enforcement – directly affects intermediaries’ rents. This creates incentives not only to sabotage a tough principal but also to collude with a lenient one. Technologically, I study an intermediary who affects electoral selection through the equilibrium level of corruption itself, rather than through a publicly observed policy or service flow. The consequence is a distinct selection problem for voters: lower observed corruption may reflect either a genuinely favorable enforcement environment or strategic collusion designed to preserve it. That mechanism generates a non-monotonic relationship between enforcement and corruption and identifies conditions under which corruption can worsen under reform-minded principals.

Model

I study a two-period game, $t \in \{1, 2\}$, between a representative Voter (V) and an unelected intermediary embedded in the bureaucracy. For tractability, I model this intermediary as a single strategic actor, the Bureaucrat (B). The political environment also contains two elected officials, an Incumbent (I), who serves in period 1, and a Challenger (C), who may replace the incumbent in period 2. Politicians are non-strategic. They differ only in their enforcement type.

At the beginning of the game, Nature draws the types of the potential politicians, $P \in \{I, C\}$. A politician’s type is her enforcement level, e^P , drawn independently from a uniform distribution,

$$e^P \sim U[\underline{e}, 1].$$

The lower bound $\underline{e} \in (0, 1)$ captures the minimum baseline of political enforcement, with the upper bound normalized to 1. A lower $\underline{e}$ corresponds to a weaker political environment, one in which even relatively lenient politicians are plausible officeholders.

In each period t , after observing the enforcement level e^P of the politician in office, the Bureaucrat chooses a level of corruption $k_t \in \mathbb{R}$.¹ The Bureaucrat’s per-period payoff is

$$u_t^B(k_t, e^P) = k_t - \alpha k_t^2 - e^P k_t^2,$$

¹Because the Bureaucrat is a reduced-form intermediary, this choice captures how much corruption he permits within the organization he supervises, whether by restraining subordinates, turning a blind eye, or allowing rent extraction to proceed.

where $\alpha > 0$ measures the state's broader institutional capacity to deter corruption. A higher α corresponds to a more robust and professional state apparatus – for example, more effective auditing bodies or a stronger rule of law – that imposes an independent additive cost on graft, distinct from the active enforcement effort of the political leader in office.

After the Bureaucrat chooses first-period corruption, k_1 , the Voter observes a noisy public signal, s_1 . The signal is additive

$$s_1 = k_1 + \varepsilon,$$

where the noise term ε is uniformly distributed on $\left[-\frac{1}{2q}, \frac{1}{2q}\right]$. The parameter $q > 0$ captures the precision of the information environment, reflecting the quality of monitoring institutions or media independence. A higher q narrows the support of the noise distribution and makes the public signal a more accurate measure of underlying corruption.

At the end of period 1, an election is held. Based on the realized signal s_1 , the Voter decides whether to retain the Incumbent ($r = 0$) or replace her with the Challenger ($r = 1$). The Voter's per-period utility is $u_t^V(k_t) = -k_t$. Although the Voter's objective is to select the politician expected to generate the lowest level of corruption in period 2, I begin by assuming that she follows a simple retrospective threshold rule: the Incumbent is replaced if and only if s_1 exceeds an exogenous cutoff $\hat{s}$. This simplifies the Bureaucrat's problem and allows me to isolate his strategic electoral incentives before endogenizing the Voter's strategy later in the paper.

In period 2, the Bureaucrat observes the enforcement level e_2 of the politician in office (either e^I or e^C) and then chooses second-period corruption, k_2 . The game then ends and payoffs are realized. The Bureaucrat and the Voter share a common discount factor $\delta \in (0, 1]$.

Non-strategic Corruption Benchmark

I begin with a static benchmark in which the Bureaucrat does not internalize the electoral consequences of corruption. This benchmark provides the reference point for the strategic distortions that follow. Let $k^M(e)$ denote the non-strategic level of corruption under enforcement level e . Formally, for any given e , $k^M(e)$ solves

$$k^M(e) \equiv \arg \max \{ \pi(k) - c(k, e) \}. \quad (1)$$

Equivalently, the optimum is characterized by

$$\pi'(k^M(e)) = c_k(k^M(e), e).$$

Under the parametric assumptions of the model, the Bureaucrat's problem admits a closed-form solution.

Lemma 1. *The non-strategic level of corruption and the associated single-period payoff are*

$$k^M(e) = \frac{1}{2(\alpha + e)} \quad \text{and} \quad u^{B,M}(e) \equiv u^B(k^M(e), e) = \frac{1}{4(\alpha + e)}.$$

Proof. See Appendix A. □

Absent electoral incentives, corruption is decreasing in both the politician’s enforcement level, e , and the state’s capacity, α . Higher enforcement raises the marginal cost of corruption through the politician’s own anti-corruption effort, while higher state capacity raises that cost through the broader institutional environment. Both forces therefore reduce the amount of corruption the Bureaucrat is willing to permit. The next proposition records these comparative statics.

- Proposition 1.** 1. *The non-strategic level of corruption, $k^M(e)$, is strictly decreasing in enforcement, e , and state capacity, α .*
2. *The expected non-strategic level of corruption, $\mathbb{E}[k^M(e)]$, is strictly decreasing in the minimum institutional baseline, $\underline{e}$.*

Proof. See Appendix B. □

Figure 1 illustrates this benchmark. For any fixed level of state capacity, stronger enforcement reduces equilibrium corruption. An increase in state capacity shifts the entire schedule downward. In the non-strategic benchmark, then, stronger institutions are uniformly disciplining. The remainder of the analysis shows how this monotone logic breaks down once the Bureaucrat internalizes the electoral consequences of corruption.

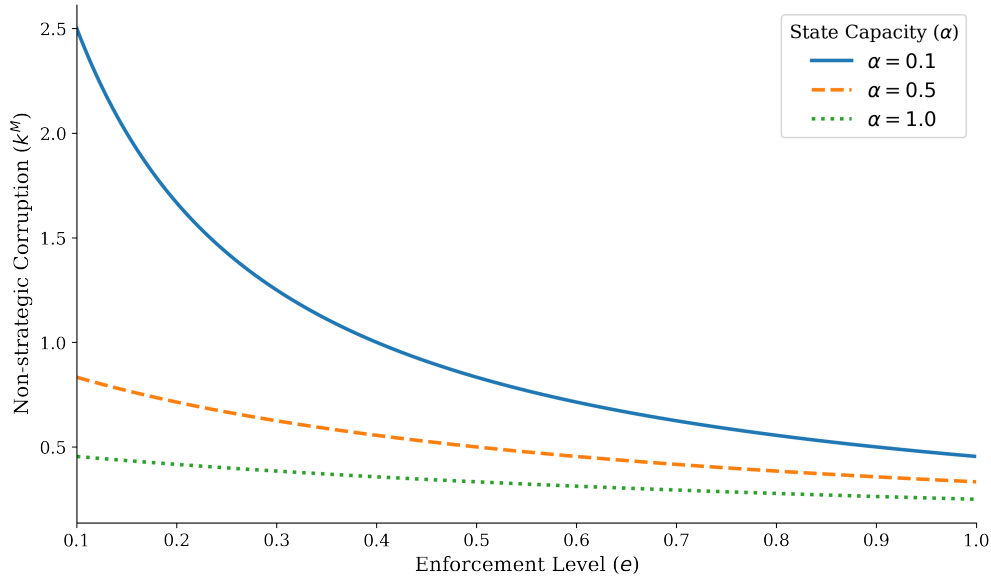


Figure 1: Non-strategic corruption as a function of enforcement.

Strategic Corruption

I now turn to the dynamic case. To isolate the Bureaucrat’s political incentives, I begin by taking the Voter’s behavior as given. Specifically, suppose the Voter follows a simple retrospective rule: she replaces the Incumbent if and only if the observed corruption signal, s_1 , exceeds a fixed cutoff, $\hat{s}$. This assumption temporarily shuts down the Voter’s inference problem and turns the model into a dynamic optimization problem for the Bureaucrat.

Given the additive signal structure,

$$s_1 = k_1 + \varepsilon, \quad \varepsilon \sim U \left[-\frac{1}{2q}, \frac{1}{2q} \right],$$

the Bureaucrat knows that choosing first-period corruption k_1 induces a probability of replacement

$$p(k_1) = \Pr(\varepsilon > \hat{s} - k_1),$$

which is given by

$$p(k_1) = \begin{cases} 0 & \text{if } \hat{s} > k_1 + \frac{1}{2q}, \\ \frac{1}{2} + q(k_1 - \hat{s}) & \text{if } \hat{s} \in \left[k_1 - \frac{1}{2q}, k_1 + \frac{1}{2q} \right], \\ 1 & \text{if } \hat{s} < k_1 - \frac{1}{2q}. \end{cases}$$

I focus on the interior region, where the cutoff lies within the support of the signal. Substantively, this is the most interesting case for the analysis: it is the only region where the Bureaucrat's choice of corruption has a marginal impact on the electoral outcome ($p'(k_1) = q > 0$). In the corner cases, the marginal probability of replacement is zero, causing the strategic incentives for sabotage and collusion to vanish and the model to revert to the non-strategic benchmark.

In period 2, the election is over and the Bureaucrat solves the static problem. Conditional on the enforcement level e_2 of the politician in office, he chooses

$$k_2^* = k^M(e_2) = \frac{1}{2(\alpha + e_2)}$$

in every equilibrium.

The period-1 problem is different because current corruption affects both current rents and future political control. Given incumbent type e^I , the Bureaucrat chooses k_1 to maximize

$$V(k_1, e^I) = k_1 - (\alpha + e^I)k_1^2 + \delta \left((1 - p(k_1))u^{B,M}(e^I) + p(k_1)V_B^C \right),$$

where $\delta \in (0, 1]$ is the discount factor, $u^{B,M}(e^I) = \frac{1}{4(\alpha + e^I)}$ is the continuation payoff if the Incumbent is retained, and $V_B^C \equiv \mathbb{E}[u^{B,M}(e^C)] = \frac{\ln(\alpha+1) - \ln(\alpha+e)}{4(1-e)}$ is the expected continuation payoff if the Incumbent is replaced by a randomly drawn Challenger.

Define

$$\Delta(e^I) \equiv u^{B,M}(e^I) - V_B^C$$

as the *retention value* of the Incumbent. The Bureaucrat's optimal first-period corruption choice is then

$$k_1^*(e^I) \equiv \frac{1 - \delta q \Delta(e^I)}{2(\alpha + e^I)} = k^M(e^I) - \frac{\delta q \Delta(e^I)}{2(\alpha + e^I)}. \quad (2)$$

Equation (2) makes the political distortion transparent. When the Incumbent is lenient, $\Delta(e^I) > 0$: retaining her is valuable to the Bureaucrat, so he lowers corruption below the static benchmark in order to reduce the probability of replacement. I refer to this as *strategic collusion*. When the Incumbent is tough, $\Delta(e^I) < 0$: the Bureaucrat prefers turnover and therefore raises corruption above the static benchmark in order to increase the probability of removal. I refer to this as *strategic sabotage*. The next proposition formalizes this distinction.

Proposition 2. *There exists a unique enforcement threshold*

$$e^* \equiv \frac{1 - \underline{e}}{\ln(\alpha + 1) - \ln(\alpha + \underline{e})} - \alpha \in (\underline{e}, 1)$$

such that:

1. if $e^I \leq e^*$, the Bureaucrat engages in strategic collusion, so that $k_1^*(e^I) \leq k^M(e^I)$;
2. if $e^I > e^*$, the Bureaucrat engages in strategic sabotage, so that $k_1^*(e^I) > k^M(e^I)$.

Proof. See Appendix C. □

The cutoff e^* separates political environments in which the Bureaucrat wants continuity from those in which he wants turnover. I next examine how that threshold varies with state capacity and the baseline quality of the political class.

Proposition 3. *The sabotage threshold e^* is strictly increasing in both state capacity, α , and the minimum institutional baseline, $\underline{e}$.*

An increase in state capacity, α , makes corruption less profitable under any politician. This compresses the payoff difference between retaining the Incumbent and drawing a random Challenger. Turnover is therefore less attractive, so sabotage arises only against sufficiently harsh incumbents. The threshold e^* rises. An increase in the minimum institutional baseline, $\underline{e}$, affects the composition of the challenger pool. When $\underline{e}$ is higher, even the most lenient feasible Challenger is relatively tough. This lowers the Bureaucrat's expected payoff from replacement, makes retention more attractive for any given incumbent, and expands the range of incumbents for whom collusion is optimal. Again, e^* rises. Figure 2 illustrates both effects.

This comparative static has a useful substantive implication. Stronger states do not necessarily eliminate political manipulation by intermediaries. Instead, they can change its form. As the overall environment becomes less permissive of rent extraction, the Bureaucrat becomes less willing to inflate corruption to unseat a tough incumbent and more willing to suppress corruption to preserve a relatively favorable one.

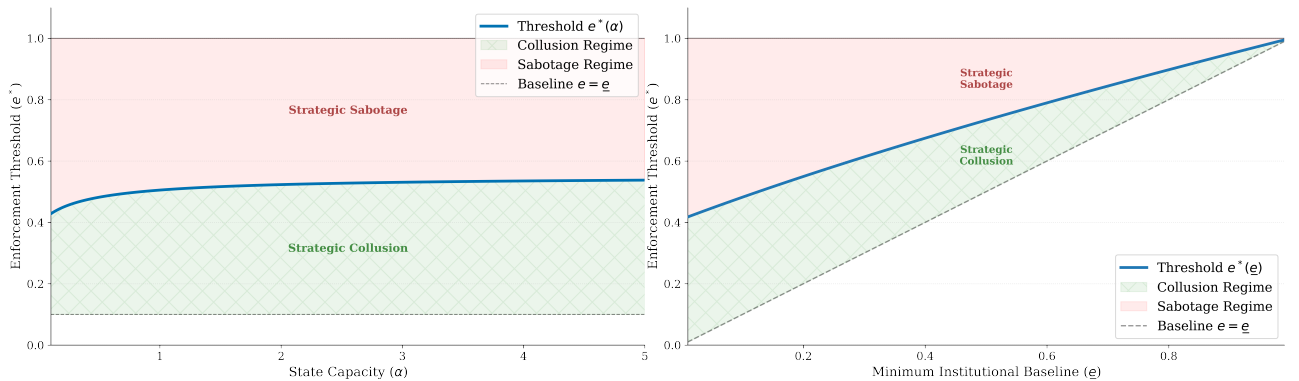


Figure 2: Effects of state capacity α (left panel) and the minimum institutional baseline $\underline{e}$ (right panel) on the threshold e^* separating collusion from sabotage.

I next consider the role of the information environment.

Proposition 4. *An increase in information precision, q , has opposite effects depending on the Incumbent's type:*

1. *when $e^I < e^*$, higher information precision strictly decreases first-period corruption, so that $\frac{\partial k_1^*}{\partial q} < 0$;*
2. *when $e^I > e^*$, higher information precision strictly increases first-period corruption, so that $\frac{\partial k_1^*}{\partial q} > 0$.*

Proof. See Appendix G. □

Information precision q determines how effective corruption is as an *electoral* instrument. Under a lenient incumbent, the Bureaucrat wants retention. A more precise signal means that marginal reductions in corruption translate more directly into electoral survival, which strengthens collusion and lowers equilibrium corruption. Under a tough incumbent, the Bureaucrat wants removal. A more precise signal means that marginal increases in corruption are more likely to trigger turnover, which strengthens sabotage and raises equilibrium corruption.

This result shows that greater transparency is not uniformly disciplining. Better auditing or greater media independence makes realized corruption more consequential for electoral survival. As a result, when intermediaries seek to unseat a reform-minded principal, a more informative signal can increase equilibrium corruption rather than reduce it. Figure 3 illustrates this comparative static. The figure plots equilibrium first-period corruption, k_1^* , against information precision, q , for two incumbent types. Under collusion, higher q lowers corruption because restraint is more effective at securing reelection. Under sabotage, higher q raises corruption because allowing more corruption becomes a more effective way to trigger incumbent replacement.

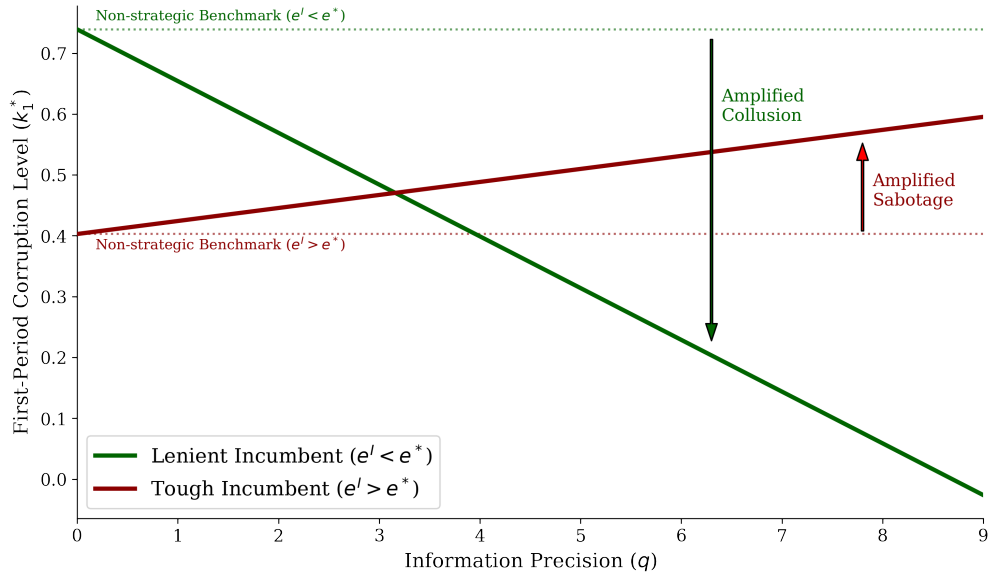


Figure 3: Effect of information precision q on first-period corruption.

I next turn to the effect of the incumbent's own enforcement level. An increase in e^I triggers two opposing forces. First, there is a *direct deterrence effect*: higher enforcement increases the marginal cost of corruption. Second, there is an *indirect political effect*: higher enforcement lowers the Bureaucrat's continuation payoff under the Incumbent and, thus, makes turnover more attractive.

Proposition 5. *Let $\bar{\alpha} \equiv \frac{\delta q}{2(1+\delta q V_B^C)} - \underline{e}$. Then:*

1. *if $\alpha < \bar{\alpha}$, there exists a unique threshold*

$$e^{**}(\alpha) \equiv \frac{\delta q}{2(1 + \delta q V_B^C)} - \alpha \in (\underline{e}, 1)$$

such that $k_1^(e^I)$ is strictly increasing in enforcement for $e^I < e^{**}(\alpha)$ and strictly decreasing for $e^I > e^{**}(\alpha)$;*

2. *if $\alpha \geq \bar{\alpha}$, then $k_1^*(e^I)$ is strictly decreasing in enforcement for all $e^I \in (\underline{e}, 1]$.*

Proof. See Appendix E. □

Proposition 5 identifies when the political motive is strong enough to overturn the usual disciplining effect of enforcement. When state capacity is low, there exists a threshold e^{**} such that corruption first rises with enforcement and falls only once enforcement becomes sufficiently strong. For $e^I < e^{**}$, a marginal increase in enforcement does little to raise the direct cost of corruption, but it does reduce the value of retaining the incumbent. In that region, the political effect dominates, and equilibrium corruption rises. For $e^I > e^{**}$, the direct cost of corruption becomes large enough to dominate, and equilibrium corruption falls. When state capacity is high, by contrast, corruption is already costly enough that the deterrence effect dominates everywhere and the upward-sloping region disappears. Figure 4 illustrates this result. The figure plots equilibrium first-period corruption, k_1^* , against the incumbent's enforcement level, e^I . When capacity is low, the schedule is hump-shaped: corruption initially rises with enforcement and then falls once enforcement becomes sufficiently strong.

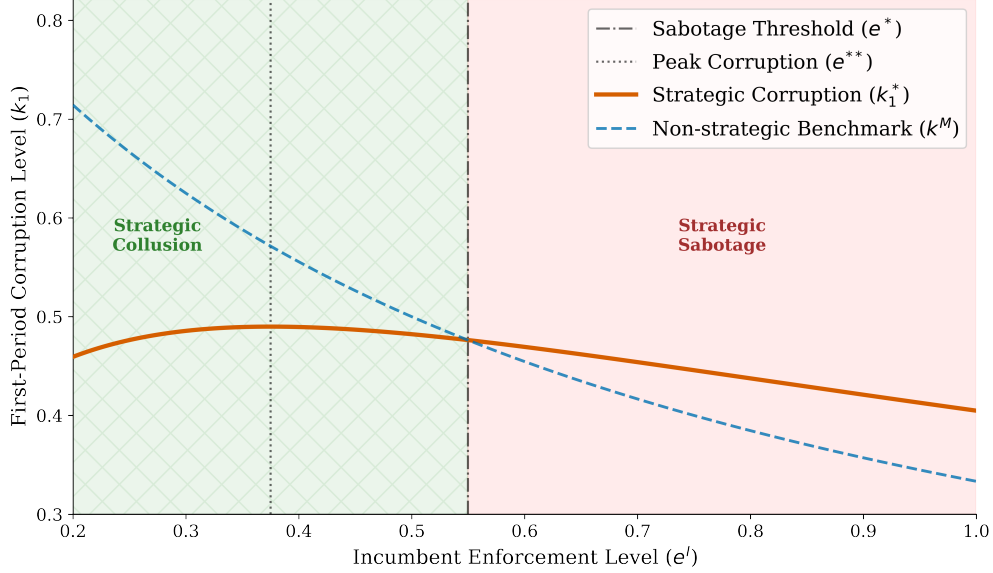


Figure 4: First-period corruption as a function of enforcement.

The next proposition turns from the incumbent's enforcement level to the broader institutional environment.

Proposition 6. *There exists a unique capacity threshold $\alpha^* \geq 0$ such that:*

1. *when $\alpha < \alpha^*$, first-period corruption is strictly increasing in state capacity, so that $\frac{\partial k_1^*}{\partial \alpha} > 0$;*
2. *when $\alpha \geq \alpha^*$, first-period corruption is strictly decreasing in state capacity, so that $\frac{\partial k_1^*}{\partial \alpha} \leq 0$.*

Proof. See Appendix F. □

The effect of state capacity is itself non-monotonic because α works through two channels. On the one hand, higher state capacity directly raises the cost of corruption and therefore discourages it. On the other hand, higher state capacity compresses continuation payoffs across politicians: when corruption is less profitable under any principal, the stakes of political turnover fall. This weakens the Bureaucrat's incentive to manipulate the election.

When state capacity is initially low, the second force can dominate. A marginal increase in α weakens the Bureaucrat's incentive to restrain corruption under favorable incumbents and can therefore move behavior back toward the static rent-maximizing benchmark, raising equilibrium corruption. Once α is sufficiently high, however, the direct deterrent effect dominates, and further increases in state capacity reduce corruption. Figure 5 illustrates this point.

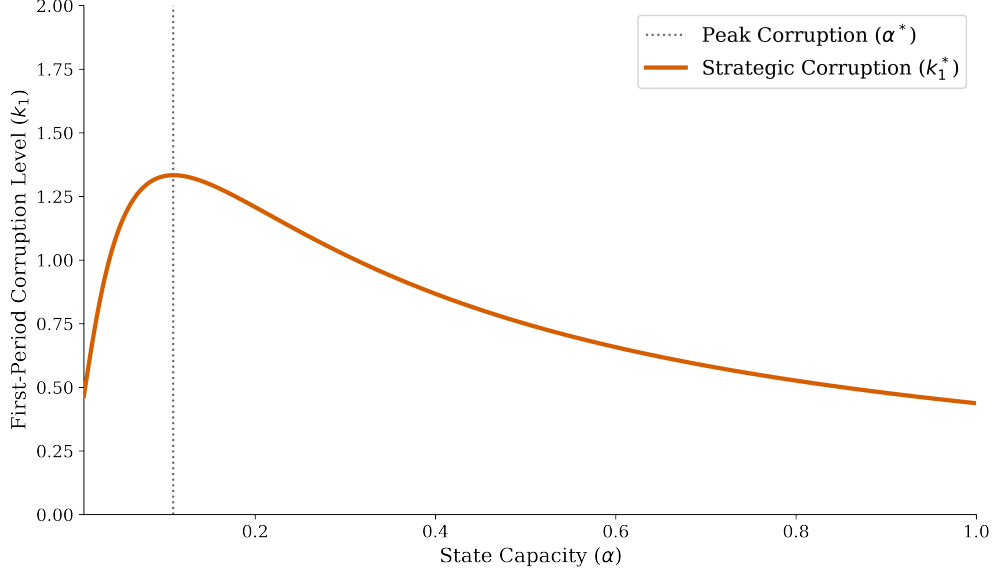


Figure 5: First-period corruption as a function of state capacity α .

Finally, note that under uniform noise, the Bureaucrat's optimal first-period corruption choice is independent of the Voter's cutoff $\hat{s}$. A marginal change in corruption shifts a constant density of signal realizations across the electoral threshold. Substantively, $\hat{s}$ determines the baseline probability of incumbent survival, and because the Bureaucrat optimizes at the margin, his strategic distortion would depend on the responsiveness of the information environment, but not on the Voter's standard for accountability.

Endogenous Electoral Accountability

Having characterized the Bureaucrat's optimal corruption choice for a fixed electoral rule, I now endogenize the Voter's decision and solve for the equilibrium of the full game. The Voter seeks to minimize second-period corruption, which is equivalent to maximizing expected second-period enforcement. After observing the public signal $s_1 = k_1^* + \varepsilon$, she updates her beliefs about the Incumbent's type using Bayes' rule and retains the Incumbent if and only if the posterior expected enforcement exceeds the expected enforcement of a random Challenger:

$$\mathbb{E}[e^I \mid s_1] \geq \mathbb{E}[e^C] = \bar{e} \equiv \frac{1 + \underline{e}}{2}, \quad (3)$$

where $\bar{e}$ denotes the enforcement level of the average politician.

Proposition 7. *Let*

$$\hat{s} \equiv k_1^*(\bar{e}) = \frac{1 - \delta q \Delta(\bar{e})}{2(\alpha + \bar{e})}.$$

Then:

1. *If $\alpha \geq \bar{\alpha}$, there exists a unique pure-strategy Perfect Bayesian Equilibrium in which the Voter retains the Incumbent if and only if $s_1 \leq \hat{s}$.*

2. If $\alpha < \bar{\alpha}$, a pure-strategy Perfect Bayesian Equilibrium exists if and only if

(i) $k_1^*(1) \leq k_1^*(\underline{e})$; and

(ii) $e^{**}(\alpha) \leq \bar{e}$.

When it exists, the equilibrium is unique, and the Voter retains the Incumbent if and only if $s_1 \leq \hat{s}$.

Proof. See Appendix H. □

Proposition 7 characterizes when conditions under which pure-strategy PBE exists. When $\alpha \geq \bar{\alpha}$, equilibrium corruption is decreasing in enforcement. Lower realized corruption is therefore always good news about the Incumbent's type. The Voter then follows a threshold rule: she retains the Incumbent after sufficiently favorable signals and replaces otherwise. The cutoff $\hat{s}$ is pinned down by the type $\bar{e}$, which makes the Voter exactly indifferent between retaining the Incumbent and drawing a random Challenger.

When $\alpha < \bar{\alpha}$, equilibrium corruption is no longer globally decreasing in enforcement. A higher-enforcement politician may sometimes generate more corruption because the Bureaucrat has a stronger incentive to induce turnover. The two additional conditions ensure that this non-monotonicity does not overturn the ordering of types relevant for the Voter's decision. Condition (i) requires that the strongest type still generate no more corruption than the weakest type. Condition (ii) requires that the peak of the corruption schedule occur among below-average incumbents, so that around the Voter's relevant margin the signal remains informative about enforcement. Under these conditions, the Voter's optimal strategy is again a cutoff rule. If either condition fails, the signal becomes too confounded to support a any pure-strategy equilibrium.

Figure 6 illustrates these equilibrium restrictions. It plots the regions in (q, α) space for which a pure-strategy Perfect Bayesian Equilibrium exists. The figure shows that pure-strategy equilibrium is guaranteed when state capacity is sufficiently high, so that corruption is monotone in enforcement. When state capacity is lower, existence depends on the precision of the information environment: only when the non-monotonicity remains limited enough does the signal preserve a useful ordering of types.

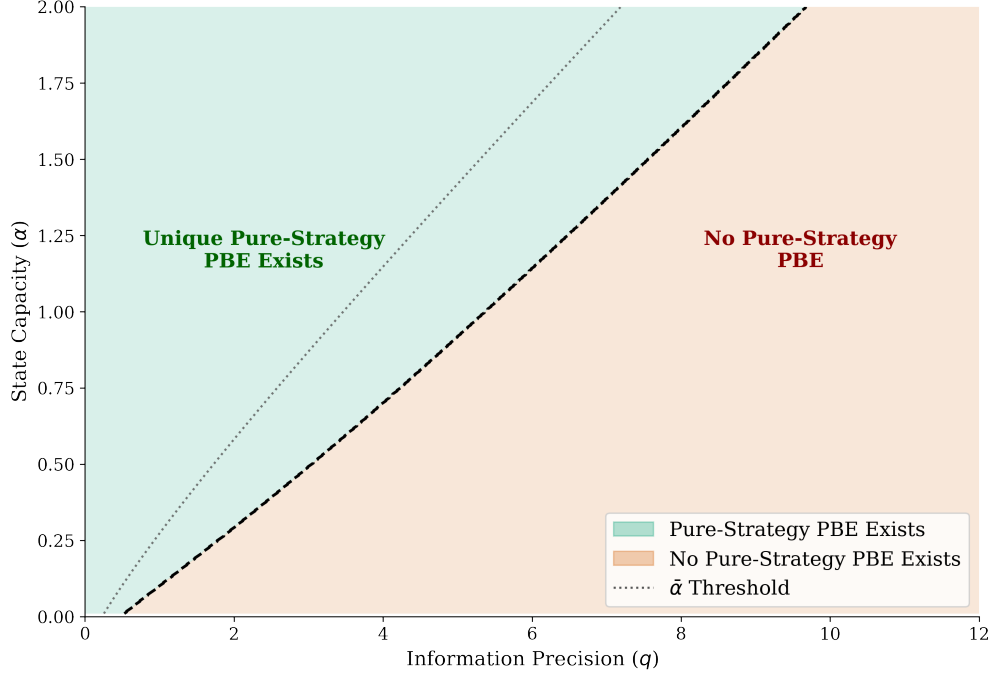


Figure 6: Regions of (q, α) for which a pure-strategy Perfect Bayesian Equilibrium exists.

Welfare Implications

I now turn from equilibrium behavior to voter welfare. In the pure-strategy equilibria characterized above, information precision, q , and state capacity, α , affect welfare through two channels. They change current-period corruption, but they also change electoral selection by affecting both the informativeness of the public signal and the Bureaucrat's incentive to manipulate corruption. The welfare comparative statics therefore need not mirror the earlier comparative statics for equilibrium corruption.

Assumption 1. *To keep the analysis in the interior region of the pure-strategy equilibrium, I assume that information precision satisfies $q < q_{\max}$, where*

$$q_{\max} \equiv \frac{\frac{1}{\alpha+\underline{e}} - \frac{1}{\alpha+\bar{e}} + \sqrt{\frac{1}{\alpha+\underline{e}} - \frac{1}{(\alpha+\bar{e})^2} + 4\delta \left(\frac{\Delta(\underline{e})}{\alpha+\underline{e}} - \frac{\Delta(\bar{e})}{\alpha+\bar{e}} \right)}}{2\delta \left(\frac{\Delta(\underline{e})}{\alpha+\underline{e}} - \frac{\Delta(\bar{e})}{\alpha+\bar{e}} \right)}.$$

Assumption 1 is a technical regularity condition. It rules out cases in which the signal is so precise that, for some types, marginal changes in corruption make replacement effectively deterministic. In that region, corruption no longer affects electoral outcomes at the margin, and the strategic sabotage–collusion mechanism shuts down. The bound on q therefore keeps the analysis on the interior part of the model, where information remains electorally relevant without eliminating manipulation.

I begin with a benchmark in which the Bureaucrat is non-strategic and does not manipulate corruption to affect electoral selection.

Proposition 8. *In the non-strategic benchmark, the Voter’s expected utility is increasing in both information precision, q , and state capacity, α .*

Proof. See Appendix TBA. □

Holding bureaucratic behavior fixed, an increase in q makes realized corruption a more informative signal of the incumbent’s type and therefore improves electoral selection. An increase in α raises the cost of corruption and lowers graft under any politician. In the benchmark, both more precise information and greater state capacity unambiguously improve voter welfare.

Once the Bureaucrat behaves strategically, however, neither comparative static remains monotone. Information quality and state capacity now affect not only corruption directly, but also the Bureaucrat’s incentive to manipulate corruption in order to influence incumbent survival.

Proposition 9. *Within the class of pure-strategy equilibria, there exists a unique threshold $q^* > 0$ such that the Voter’s expected utility is increasing in information precision, q , for $q < q^*$ and weakly decreasing in q for $q \geq q^*$.*

Proof. See Appendix I. □

A marginal increase in precision makes the public signal more informative about the underlying level of corruption and, thus, other things being equal should improve voter welfare. However, the greater precision also strengthens the Bureaucrat’s incentive to manipulate the signal: under a lenient incumbent, the Bureaucrat has stronger incentives to suppress corruption to protect a favorable principal, under a tough incumbent, he has stronger incentives to inflate corruption to induce turnover. As a result, the Voter may become better able to read the signal, but the signal itself becomes less informative about the object she cares about. At low levels of q , the gain in inference dominates. At high levels of q , strategic manipulation undermines electoral selection, and the voter welfare falls.²

Figure 7 illustrates this result. It plots the Voter’s expected utility against information precision, q , within the pure-strategy equilibrium region under strategic and non-strategic bureaucrat. For the non-strategic bureaucrat the voter welfare improves in q . For the strategic bureaucrat the voter welfare is initially upward sloping: better information improves selection and raises welfare. Beyond q^* , it turns downward as the Bureaucrat’s strategic response overwhelms the direct informational gain.

²Note that Assumption 1 places an upper bound on information precision. Beyond that threshold, precision no longer strengthen the Bureaucrat’s incentive to manipulate the signal and voter welfare is increasing in q .

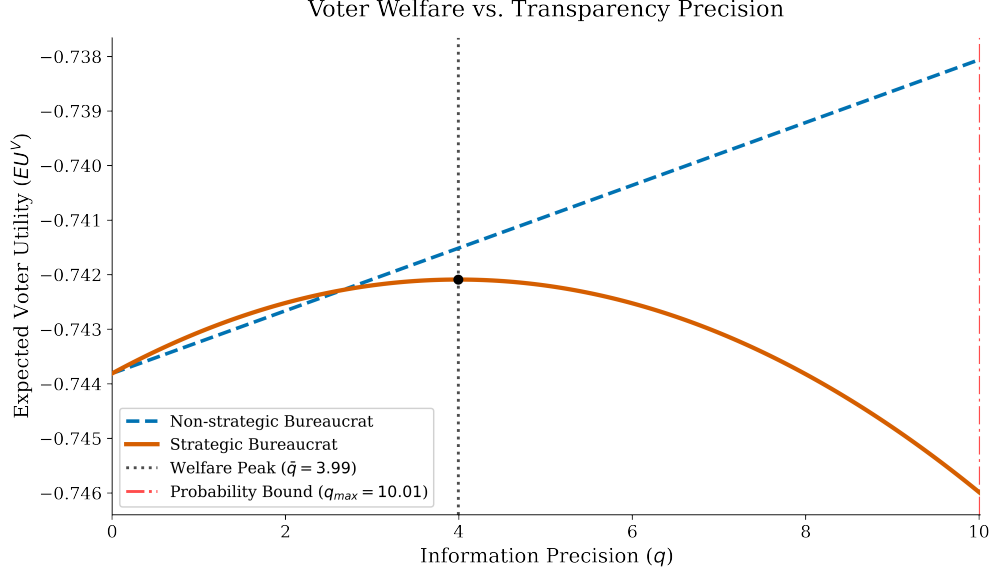


Figure 7: Voter expected utility as a function of information precision q in pure-strategy equilibrium.

The next proposition turns to the effect of state capacity.

Proposition 10. *Within the class of pure-strategy equilibria, there exists a unique threshold $\alpha^V \geq 0$ such that the Voter's expected utility is decreasing in state capacity for $\alpha < \alpha^V$ and weakly increasing in α for $\alpha \geq \alpha^V$.*

Proof. See Appendix I. □

Higher state capacity lowers corruption and decreases Bureaucrat's incentives to manipulate electoral outcomes, improving voter welfare. But it also reduces the stakes of electoral selection. When state capacity is low, politician types matter greatly: replacing a lenient incumbent with a tougher one implies a large drop in second-period corruption. As α increases the gap between lenient and tough types narrows. At low levels of α , this compression in the value of selecting the right type can dominate the direct reduction in corruption and the improvement in the signal, so voter welfare falls. Once α is sufficiently high, however, the direct anti-corruption effect dominates, and further increases in state capacity raise voter welfare.

Figure 8 illustrates this comparative static. It plots the Voter's expected utility against state capacity, α under strategic and non-strategic bureaucrat. With non-strategic bureaucrat the welfare improves in α , with the strategic one, the welfare is initially downward sloping, reflecting the loss in the value of electoral selection, and then upward sloping once the direct anti-corruption effect of state capacity dominates.

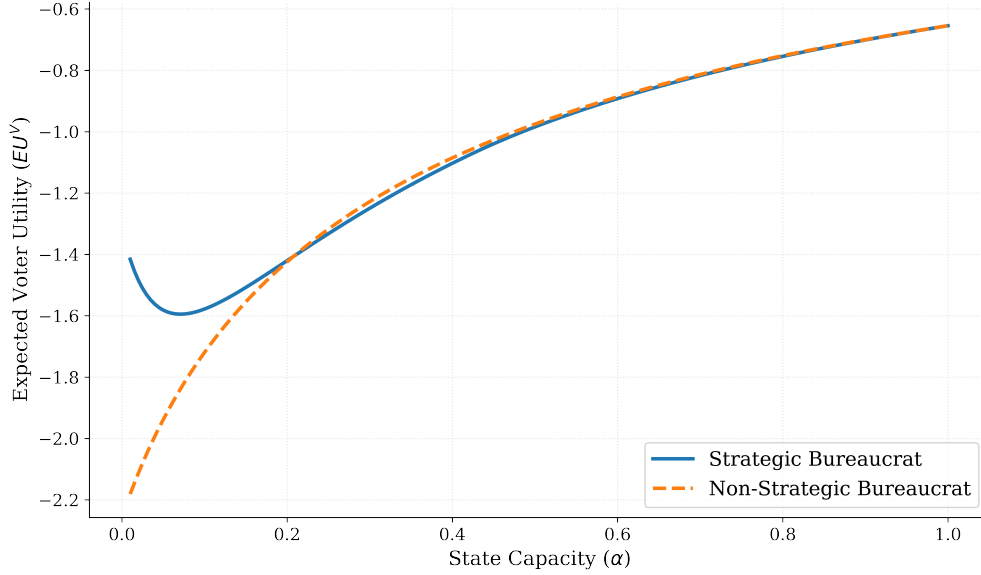


Figure 8: Voter expected utility as a function of state capacity α in pure-strategy equilibrium.

Discussion

The model qualifies a standard implication of principal-agent accounts of corruption and oversight. In those settings, stronger monitoring and sharper performance measures discipline agents by increasing the expected cost of misconduct. In this paper, oversight has an additional effect: it changes the intermediary's continuation value by affecting which principal is likely to remain in office. Once that selection margin is introduced, the comparative statics need not be monotone. A tougher principal may initially induce more corruption, not less, because a hostile intermediary has reason to permit greater misconduct in order to increase the odds of turnover. Only when enforcement becomes sufficiently strong does the direct deterrent effect dominate. Information quality generates a similar tension. Better signals improve accountability, but they also make corruption a more effective instrument for influencing political survival.

The welfare results sharpen this point. While both better information and higher state capacity increase voter welfare under non-strategic benchmark, under strategic bureaucrat neither result is guaranteed. Greater information precision improves the Voter's inference, but it also strengthens the Bureaucrat's incentive to manipulate corruption for electoral purposes. Greater state capacity lowers corruption under any politician, but it also compresses the difference between politician types and therefore reduces the value of electoral selection. When capacity is low, that loss in selection value can dominate the direct anti-corruption effect. More generally, the results suggest that institutional reforms should be evaluated not only by how they change corruption directly, but also by how they reshape the incentives of intermediaries and the returns to accountability.

Although the model is motivated by anti-corruption, the mechanism is not specific to politics. In firms and other organizations, middle managers often control whether subordinate misconduct is contained or allowed to persist, while the resulting outcomes shape

the evaluation, retention, or reassignment of their superiors. Classic work in organizational economics emphasizes how imperfect performance measures distort effort, induce gaming, and shape the allocation of authority . The present model adds a dynamic selection channel. Intermediaries may tolerate or restrain misconduct not only because of its current payoff consequences, but also because it affects who will supervise them tomorrow. That link places the paper in conversation with recent work on sabotage and strategic policy degradation . The distinctive implication here is that the same intermediary may collude with a lenient principal and sabotage a strict one.

Appendix

A Lemma 1

Let the Bureaucrat's objective function be given by $u^B(k, e) = k - \alpha k^2 - ek^2$. The first-order condition for an interior maximum with respect to corruption k is

$$\frac{\partial u^B}{\partial k} = 1 - 2\alpha k - 2ek = 1 - 2(\alpha + e)k = 0.$$

Solving for k we get $k^M(e) = \frac{1}{2(\alpha + e)}$. To verify this is a global maximum, evaluate the second-order condition

$$\frac{\partial^2 u^B}{\partial k^2} = -2(\alpha + e).$$

Given $\alpha > 0$ and $e > 0$, the SOC is strictly negative, function is globally concave. To find single-period payoff, substitute $k^M(e)$ back into the utility function

$$u^{B,M}(e) = \left(\frac{1}{2(\alpha + e)} \right) - (\alpha + e) \left(\frac{1}{2(\alpha + e)} \right)^2 = \frac{1}{4(\alpha + e)}.$$

B Proposition 1

For part 1, differentiating the non-strategic corruption baseline $k^M(e) = \frac{1}{2(\alpha + e)}$ with respect to e and α , we get:

$$\frac{\partial k^M(e)}{\partial e} = -\frac{1}{2(\alpha + e)^2} < 0 \quad \text{and} \quad \frac{\partial k^M(e)}{\partial \alpha} = -\frac{1}{2(\alpha + e)^2} < 0.$$

Both derivatives are strictly negative since $\alpha > 0$ and $e \geq \underline{e} > 0$.

For part 2, the expected non-strategic corruption level is

$$\mathbb{E}[k^M(e)] = \int_{\underline{e}}^1 \left(\frac{1}{2(\alpha + e)} \right) \frac{1}{1 - \underline{e}} de = \frac{\ln(\alpha + 1) - \ln(\alpha + \underline{e})}{2(1 - \underline{e})}.$$

Differentiate it with respect to $\underline{e}$

$$\frac{\partial \mathbb{E}[k^M(e)]}{\partial \underline{e}} = \frac{\ln(\alpha + 1) - \ln(\alpha + \underline{e}) - \frac{1 - \underline{e}}{\alpha + \underline{e}}}{2(1 - \underline{e})^2}.$$

Let $g(\underline{e}) = \ln(\alpha + 1) - \ln(\alpha + \underline{e}) - \frac{1-\underline{e}}{\alpha + \underline{e}}$. Note that at the upper bound, $g(1) = 0$. Differentiate $g(\underline{e})$ with respect to $\underline{e}$

$$g'(\underline{e}) = -\frac{1}{\alpha + \underline{e}} - \left(\frac{-1(\alpha + \underline{e}) - (1 - \underline{e})(1)}{(\alpha + \underline{e})^2} \right) = -\frac{1}{\alpha + \underline{e}} + \frac{\alpha + 1}{(\alpha + \underline{e})^2} = \frac{1 - \underline{e}}{(\alpha + \underline{e})^2} > 0.$$

Thus, $g(\underline{e})$ is negative for all $\underline{e} \in (0, 1)$. Therefore, $\frac{\partial \mathbb{E}[k^M(e)]}{\partial \underline{e}} < 0$.

C Proposition 2

The retention value $\Delta(e^I) = 0$ if and only if

$$\frac{1}{4(\alpha + e^I)} = \frac{\ln(\alpha + 1) - \ln(\alpha + \underline{e})}{4(1 - \underline{e})}.$$

Solving for e^I we get the threshold $e^* = \frac{1-\underline{e}}{\ln(\alpha+1)-\ln(\alpha+\underline{e})} - \alpha$. If $e^I < e^*$, the incumbent is lenient and the retention value is strictly positive, $\Delta(e^I) > 0$. Evaluating the first-order condition at the non-strategic benchmark $k^M(e^I) = \frac{1}{2(\alpha+e^I)}$, we get

$$F(k^M, e^I) = -\delta p'(k^M) \Delta(e^I) < 0.$$

Given global concavity, the first derivative is decreasing and the optimal choice must be strictly lower than the benchmark $k_1^*(e^I) < k^M(e^I)$. Conversely, if $e^I > e^*$, the retention value is negative, $\Delta(e^I) < 0$. Evaluating the FOC at k^M we get

$$F(k^M, e^I) = -\delta p'(k^M) \Delta(e^I) > 0.$$

By global concavity, the optimal level of corruption must be strictly higher $k_1^*(e^I) > k^M(e^I)$.

D Proposition 3

Note that

$$\frac{\partial(\ln(\alpha + 1) - \ln(\alpha + \underline{e}))}{\partial \alpha} = \frac{1}{\alpha + 1} - \frac{1}{\alpha + \underline{e}} = \frac{-(1 - \underline{e})}{(\alpha + 1)(\alpha + \underline{e})}$$

and, thus,

$$\frac{\partial e^*}{\partial \alpha} = \frac{(1 - \underline{e})^2}{(\ln(\alpha + 1) - \ln(\alpha + \underline{e}))^2 (\alpha + 1)(\alpha + \underline{e})} - 1.$$

Note next that $\alpha + 1 > \alpha + \underline{e} > 0$, $\alpha + 1 - (\alpha + \underline{e}) = 1 - \underline{e}$, and rewrite the derivative as

$$\frac{((\alpha + 1) - (\alpha + \underline{e}))^2}{(\ln(\alpha + 1) - \ln(\alpha + \underline{e}))^2 (\alpha + 1)(\alpha + \underline{e})} = \frac{1}{(\alpha + 1)(\alpha + \underline{e})} \left(\frac{(\alpha + 1) - (\alpha + \underline{e})}{\ln(\alpha + 1) - \ln(\alpha + \underline{e})} \right)^2.$$

For any $(\alpha + 1) \neq (\alpha + \underline{e})$, we have $\frac{(\alpha+1)-(\alpha+\underline{e})}{\ln(\alpha+1)-\ln(\alpha+\underline{e})} > \sqrt{(\alpha + 1)(\alpha + \underline{e})}$. This is equivalent to,

$\frac{1}{(\alpha+1)(\alpha+\underline{e})} \left(\frac{(\alpha+1)-(\alpha+\underline{e})}{\ln(\alpha+1)-\ln(\alpha+\underline{e})} \right)^2 > 1$. Substituting this back we get $\frac{\partial e^*}{\partial \alpha} > 0$.

Next note

$$\frac{\partial e^*}{\partial \underline{e}} = \frac{(-1)(\ln(\alpha + 1) - \ln(\alpha + \underline{e})) - (1 - \underline{e}) \left(-\frac{1}{\alpha + \underline{e}} \right)}{(\ln(\alpha + 1) - \ln(\alpha + \underline{e}))^2} = \frac{\frac{1 - \underline{e}}{\alpha + \underline{e}} - (\ln(\alpha + 1) - \ln(\alpha + \underline{e}))}{(\ln(\alpha + 1) - \ln(\alpha + \underline{e}))^2}$$

Denote the numerator by $h(\underline{e})$. Note that

$$h(1) = 0 - (\ln(\alpha + 1) - \ln(\alpha + 1)) = 0.$$

Differentiate $h(\underline{e})$ with respect to $\underline{e}$

$$h'(\underline{e}) = \frac{(-1)(\alpha + \underline{e}) - (1 - \underline{e})(1)}{(\alpha + \underline{e})^2} - \left(-\frac{1}{\alpha + \underline{e}} \right) = \frac{-(\alpha + 1)}{(\alpha + \underline{e})^2} + \frac{\alpha + \underline{e}}{(\alpha + \underline{e})^2} = -\frac{1 - \underline{e}}{(\alpha + \underline{e})^2} < 0.$$

Thus, $h(\underline{e}) > 0$ for all $\underline{e} \in (0, 1)$. Therefore, $\frac{\partial e^*}{\partial \underline{e}} > 0$.

E Proposition 5

The equilibrium level of first-period corruption is

$$k_1^*(e^I) = \frac{1 - \delta q \Delta(e^I)}{2(\alpha + e^I)} = \frac{1 + \delta q V_B^C}{2(\alpha + e^I)} - \frac{\delta q}{8(\alpha + e^I)^2}.$$

The equilibrium corruption increases in the Incumbent's enforcement when

$$\frac{\partial k_1^*}{\partial e^I} = -\frac{1 + \delta q V_B^C}{2(\alpha + e^I)^2} + \frac{\delta q}{4(\alpha + e^I)^3} > 0$$

and decreases in the enforcement otherwise. Therefore, k_1^* is increasing in e^I when

$$\frac{1 + \delta q V_B^C}{2(\alpha + e^I)^2} < \frac{\delta q}{4(\alpha + e^I)^3} \Leftrightarrow 2(\alpha + e^I)(1 + \delta q V_B^C) < \delta q \Leftrightarrow e^I < e^{**}(\alpha) \equiv \frac{\delta q}{2(1 + \delta q V_B^C)} - \alpha.$$

Finally, let us characterize conditions under which $e^{**}(\alpha) \in [\underline{e}, 1]$.

$$\frac{\delta q}{2(1 + \delta q V_B^C)} - \alpha > \underline{e} \Leftrightarrow \alpha < \bar{\alpha} \equiv \frac{\delta q}{2(1 + \delta q V_B^C)} - \underline{e}.$$

Define this boundary as $\bar{\alpha} \equiv \frac{\delta q}{2(1 + \delta q V_B^C)} - \underline{e}$. Conversely, if $\alpha \geq \bar{\alpha}$, then $e^{**}(\alpha) \leq \underline{e}$. In this case, for all possible incumbent types $e^I \in (\underline{e}, 1]$, we have $e^I > e^{**}(\alpha)$, and $\frac{\partial k_1^*}{\partial e^I} < 0$.

F Proposition 6

Differentiate $k_1^*(e^I)$ with respect to α :

$$\frac{\partial k_1^*}{\partial \alpha} = \frac{-\delta q \frac{\partial \Delta}{\partial \alpha} 2(\alpha + e^I) - 2(1 - \delta q \Delta(e^I))}{4(\alpha + e^I)^2}, \quad \frac{\partial \Delta}{\partial \alpha} = -\frac{1}{4(\alpha + e^I)^2} - \frac{\partial V_B^C}{\partial \alpha}.$$

Substituting gives

$$\frac{\partial k_1^*}{\partial \alpha} = \frac{1}{2(\alpha + e^I)^2} \left[\frac{\delta q}{2(\alpha + e^I)} - 1 - \delta q V_B^C + \delta q(\alpha + e^I) \frac{\partial V_B^C}{\partial \alpha} \right]. \quad (4)$$

Thus, the sign of $\partial k_1^*/\partial \alpha$ is the sign of

$$J(\alpha) = \delta q \left(\frac{1}{2(\alpha + e^I)} - \frac{\ln(\alpha + 1) - \ln(\alpha + \underline{e})}{4(1 - \underline{e})} - \frac{\alpha + e^I}{4(\alpha + 1)(\alpha + \underline{e})} \right) - 1.$$

Now,

$$\lim_{\alpha \rightarrow \infty} J(\alpha) = -1 < 0,$$

and

$$\lim_{\alpha \rightarrow 0} J(\alpha) = \delta q \left(\frac{1}{2e^I} + \frac{\ln(\underline{e})}{4(1 - \underline{e})} - \frac{e^I}{4\underline{e}} \right) - 1.$$

For all $\underline{e} \in (0, 1)$, the term in parentheses is positive, so there exist sufficiently large δq s.t. $\lim_{\alpha \rightarrow 0} J(\alpha) > 0$. Thus there is a range of parameters where $J(\alpha)$ crosses zero at least once.

To show uniqueness, differentiate

$$J'(\alpha) = \delta q \left(-\frac{1}{2(\alpha + e^I)^2} + (\alpha + e^I) \frac{\partial^2 V_B^C}{\partial \alpha^2} \right) = \delta q \left(-\frac{1}{2(\alpha + e^I)^2} + (\alpha + e^I) \frac{1 + 2\alpha + \underline{e}}{4(\alpha + 1)^2(\alpha + \underline{e})^2} \right).$$

At any crossing point, $J(\alpha) = 0$, so

$$V_B^C = \frac{1}{2(\alpha + e^I)} - \frac{1}{\delta q} - \frac{\alpha + e^I}{4(\alpha + 1)(\alpha + \underline{e})}.$$

Substituting this into $J'(\alpha)$ we get

$$J'(\alpha) = \frac{\delta q}{\alpha + e^I} \left(\frac{(\alpha + e^I)^3 + (\alpha + e^I)(1 - e^I)(e^I - \underline{e})}{4(\alpha + 1)^2(\alpha + \underline{e})^2} + \frac{\alpha + e^I}{4(\alpha + 1)(\alpha + \underline{e})} - \frac{1}{2(\alpha + e^I)} \right).$$

Because $e^I \in [\underline{e}, 1]$, the positive terms are bounded above by their value at $e^I = 1$, so

$$\frac{(\alpha + e^I)^3 + (\alpha + e^I)(1 - e^I)(e^I - \underline{e})}{4(\alpha + 1)^2(\alpha + \underline{e})^2} + \frac{\alpha + e^I}{4(\alpha + 1)(\alpha + \underline{e})} \leq \frac{(\alpha + 1) + (\alpha + \underline{e})}{4(\alpha + \underline{e})^2} \leq \frac{1}{2(\alpha + e^I)}$$

thus $J'(\alpha) < 0$ at every root.

Therefore if for a range of parameters there is a crossing, it is a downward crossing. Since $J(\alpha) > 0$ for α close to 0 and $J(\alpha) < 0$ for large α , there is a unique cutoff α^* such that

$$\frac{\partial k_1^*}{\partial \alpha} > 0 \quad \text{for } \alpha < \alpha^*, \quad \frac{\partial k_1^*}{\partial \alpha} < 0 \quad \text{for } \alpha > \alpha^*.$$

Thus $k_1^*(e^I)$ is increasing in α for low α and decreasing in α for high α . If for a given range of parameters there is no crossing, $k_1^*(e^I)$ is decreasing in α .

G Proposition 4

Recall that

$$k_1^*(e^I) = \frac{1 - \delta q \Delta(e^I)}{2(\alpha + e^I)} \Rightarrow \frac{\partial k_1^*}{\partial q} = -\frac{\delta \Delta(e^I)}{2(\alpha + e^I)}.$$

Given $\delta > 0$, $\alpha > 0$, and $e^I \geq \underline{e} > 0$

$$\text{sgn}\left(\frac{\partial k_1^*}{\partial q}\right) = -\text{sgn}(\Delta(e^I)).$$

From Proposition 2, we know that e^* is the unique enforcement threshold at which $\Delta(e^*) = 0$. Therefore, $\frac{\partial k_1^*}{\partial q} < 0$ when $e^I < e^*$ and $\frac{\partial k_1^*}{\partial q} \geq 0$ otherwise.

H Proposition 7

Let the Voter's pure strategy be a retention rule $r : \mathbb{R} \rightarrow \{0, 1\}$, where $r(s_1) = 0$ denotes retaining the Incumbent and $r(s_1) = 1$ denotes replacing them. The Voter strictly prefers to retain the Incumbent if and only if the posterior expected enforcement type exceeds the expected Challenger type

$$\mathbb{E}[e^I \mid s_1] \geq \mathbb{E}[e^C] = \bar{e} \equiv \frac{1 + \underline{e}}{2}. \quad (5)$$

Let $\mathcal{R} = \{s_1 \in \mathbb{R} \mid \mathbb{E}[e^I \mid s_1] \geq \bar{e}\}$ denote the equilibrium retention set.

Given the Bureaucrat's equilibrium strategy $k_1^*(e^I)$ and the uniform noise $\varepsilon \sim U\left[-\frac{1}{2q}, \frac{1}{2q}\right]$, the likelihood of observing signal s_1 given type e^I is $f(s_1 \mid e^I) = q \cdot \mathbb{1}_{\{|s_1 - k_1^*(e^I)| \leq \frac{1}{2q}\}}$. Because the prior distribution of types is uniform, Bayes' rule implies that the posterior belief $\mu(e^I \mid s_1)$ is a uniform distribution over the set of feasible types $E(s_1)$ that could have generated the signal:

$$E(s_1) = \left\{ e \in [\underline{e}, 1] \mid s_1 - \frac{1}{2q} \leq k_1^*(e) \leq s_1 + \frac{1}{2q} \right\}.$$

Because this posterior distribution is uniform, the posterior expectation is the continuous average over the set of feasible types

$$\mathbb{E}[e^I \mid s_1] = \frac{\int_{E(s_1)} e \, de}{\int_{E(s_1)} 1 \, de}.$$

Suppose first that $\alpha \geq \bar{\alpha}$. By Proposition 5, $k_1^*(e^I)$ is strictly decreasing in e^I . Consequently, its inverse $(k_1^*)^{-1}(\cdot)$ is well-defined and strictly decreasing. The set of feasible types forms a connected interval:

$$E(s_1) = \left[\max \left\{ \underline{e}, (k_1^*)^{-1} \left(s_1 + \frac{1}{2q} \right) \right\}, \min \left\{ 1, (k_1^*)^{-1} \left(s_1 - \frac{1}{2q} \right) \right\} \right].$$

Because $(k_1^*)^{-1}$ is strictly decreasing, as the observed signal s_1 increases, both the upper and lower bounds of $E(s_1)$ strictly (unless constrained by the state space boundaries $\underline{e}$ or 1) decrease. Thus, the interval $E(s_1)$ shifts downward, implying that the Voter's posterior expectation, $\mathbb{E}[e^I \mid s_1]$, is strictly decreasing in s_1 .

Because $\mathbb{E}[e^I \mid s_1]$ is monotonically decreasing, the retention condition $\mathbb{E}[e^I \mid s_1] \geq \bar{e}$ is satisfied if and only if $s_1 \leq \hat{s}$ for some unique threshold $\hat{s}$. Given $\mathbb{E}[e^I \mid k_1^*(\bar{e})] = \bar{e}$ (by the symmetry of the uniform noise), the equilibrium threshold is $\hat{s} \equiv k_1^*(\bar{e})$.

Suppose next that $\alpha < \bar{\alpha}$. By Proposition 5, when $\alpha < \bar{\alpha}$, $k_1^*(e^I)$ is non-monotonic: it strictly increases due to strategic sabotage on $[\underline{e}, e^{**}]$ and strictly decreases due to direct deterrence on $[e^{**}, 1]$. The set of feasible types might not form a connected interval. For a threshold strategy to be incentive compatible for the Voter, $\mathbb{E}[e^I \mid s_1]$ must cross $\bar{e}$ exactly once from above. This requires that the lowest possible signals are generated with high-enforcement types ($e^I > \bar{e}$), and the highest possible signals are generated with low-enforcement types ($e^I < \bar{e}$).

- (i) To ensure the lowest signals map strictly to the highest types, the global minimum of the corruption function must occur at the highest enforcement type, $e^I = 1$. Because k_1^* is single-peaked, the global minimum is either at the lower bound $\underline{e}$ or the upper bound 1. Thus, we require $k_1^*(1) \leq k_1^*(\underline{e})$.
- (ii) To ensure the highest signals unambiguously map to lower-than-average types (triggering electoral replacement), the peak of the corruption function must occur at a type below the mean. Thus, we require $e^{**}(\alpha) \leq \bar{e}$.

When both (i) and (ii) hold, as s_1 increases from its minimum, the composition of $E(s_1)$ shifts monotonically away from the right tail of the distribution (types near 1) and concentrates toward the peak around $e^{**} < \bar{e}$. Thus, $\mathbb{E}[e^I \mid s_1]$ is monotonically decreasing in the relevant neighborhood of $\bar{e}$, crossing it exactly once and sustaining the threshold rule $\hat{s} \equiv k_1^*(\bar{e})$.

Finally, if either $k_1^*(1) > k_1^*(\underline{e})$ or $e^{**}(\alpha) > \bar{e}$, there exists no pure-strategy Perfect Bayesian Equilibrium. We proceed by contradiction. Suppose there exists a pure-strategy PBE when $\alpha < \bar{\alpha}$ and either condition (i) or (ii) is violated. Let $\mathcal{R} \subseteq \mathbb{R}$ denote the Voter's equilibrium retention set, and let $k_1^*(e^I)$ denote the Bureaucrat's pure-strategy best response.

First, suppose condition (ii) fails, such that $e^{**}(\alpha) > \bar{e}$. Let $k_{max} \equiv k_1^*(e^{**})$ denote the global maximum of the corruption function. Then, for some ε there exists, $S_\varepsilon = [k_{max} + \frac{1}{2q} - \varepsilon, k_{max} + \frac{1}{2q}]$, such that $S_\varepsilon \subset \mathcal{R}$. Consider the Bureaucrat's optimal behavior when facing a lenient incumbent of type $e^I \in [\underline{e}, e^*]$. For such types, the retention value is strictly positive ($\Delta(e^I) > 0$). When such Bureaucrat deviates to k_{max} , it increases both short-term benefit and retention probability. However, given all types $e^I \leq e^*$ choose k_{max} , the Voter's posterior expectation for these signals should be below $e^* < \bar{e}$, violating the condition that $S_\varepsilon \subset \mathcal{R}$ and thereby unraveling the proposed equilibrium.

The same logic applies when condition (i) fails, such that $k_1^*(1) > k_1^*(\underline{e})$. Signaling higher corruption, then, increases retention probability. The action generating higher corruption signals becomes at least as attractive to bureaucrat with lenient types as to those with the high-enforcement types and bureaucrats with lenient types deviate to mimic upward.

I Proposition 9

The Voter's per-period utility is the negative of the corruption level, $u_t^V(k_t) = -k_t$. In the first period, $k_1^*(e^I) = \frac{1-\delta q \Delta(e^I)}{2(\alpha+e^I)}$. In the second period, if $s_1 \leq \hat{s} = \frac{1-\delta q \Delta(\bar{e})}{2(\alpha+\bar{e})}$, $k_2 = \frac{1}{2(\alpha+e^I)}$. If $s_1 > \hat{s} = \frac{1-\delta q \Delta(\bar{e})}{2(\alpha+\bar{e})}$, $k_2 = \mathbb{E}[k^M(e^C)] = \frac{\ln(\alpha+1)-\ln(\alpha+\underline{e})}{2(1-\underline{e})}$, $\bar{e}$ is defined in Equation 5.

The probability to retain the incumbent of type e^I is

$$\Pr(s_1 \leq \hat{s} \mid e^I) = \frac{1}{2} + q(\hat{s} - k_1^*(e^I)).$$

Thus

$$\begin{aligned} EU^V(e^I) &= -k_1^*(e^I) - \delta [\Pr(s_1 \leq \hat{s} \mid e^I) k^M(e^I) + (1 - \Pr(s_1 \leq \hat{s} \mid e^I)) \mathbb{E}[k^M(e^C)]] \\ &= -k_1^*(e^I) - \delta \left[\left(\frac{1}{2} + q(\hat{s} - k_1^*(e^I)) \right) \left(\frac{1}{2(\alpha+e^I)} \right) + \left(\frac{1}{2} - q(\hat{s} - k_1^*(e^I)) \right) \mathbb{E}[k^M(e^C)] \right]. \end{aligned}$$

To find the ex-ante expected utility, integrate $EU^V(e^I)$ over the prior distribution $e^I \sim U[\underline{e}, 1]$. To condense notation, define $I_k \equiv \mathbb{E} \left[\frac{1}{(\alpha+e^I)^k} \right]$, then

$$I_1 = \frac{\ln(\alpha+1) - \ln(\alpha+\underline{e})}{1-\underline{e}}, \quad I_2 = \frac{1}{1-\underline{e}} \left(\frac{1}{\alpha+\underline{e}} - \frac{1}{\alpha+1} \right), \quad I_3 = \frac{1}{2(1-\underline{e})} \left(\frac{1}{(\alpha+\underline{e})^2} - \frac{1}{(\alpha+1)^2} \right).$$

Then

$$\begin{aligned} EU^V &= - \left(\frac{1 + \delta q I_1/4}{2} I_1 - \frac{\delta q}{8} I_2 \right) \\ &\quad - \delta \left(\frac{1}{2} I_1 \left(\frac{1}{2} - q \hat{s} \right) + \left(\frac{\frac{1}{2} + q \hat{s}}{2} + \frac{q(1 + \delta q I_1/4)}{4} I_1 \right) I_1 \right. \\ &\quad \left. - \left(\frac{q(1 + \delta q I_1/4)}{4} + \frac{\delta q^2 I_1}{16} \right) I_2 + \left(\frac{\delta q^2}{16} \right) I_3 \right) \\ &= - \frac{1 + \delta}{2} I_1 + \frac{3\delta q}{8} (I_2 - I_1^2) - \frac{\delta^2 q^2}{16} (I_3 - 2I_1 I_2 + I_1^3). \end{aligned} \tag{6}$$

We now analyze effect of information precision, q ,

$$\begin{aligned} \frac{\partial EU^V}{\partial q} &= \frac{3\delta}{8} (I_2 - I_1^2) - \frac{\delta^2 q}{8} (I_3 - 2I_1 I_2 + I_1^3) \\ &= \frac{\delta}{16(1-\underline{e})^3} (\beta_0 + \beta_1 \cdot \delta q), \end{aligned} \tag{7}$$

where

$$\beta_0 = \frac{6(1-\underline{e})^3}{(1+\alpha)(\alpha+\underline{e})} - 6(1-\underline{e})(\ln(1+\alpha) - \ln(\alpha+\underline{e}))^2$$

and

$$\beta_1 = - \frac{(1-\underline{e})^3(1+2\alpha+\underline{e})}{(1+\alpha)^2(\alpha+\underline{e})^2} + \frac{4(1-\underline{e})^2}{(1+\alpha)(\alpha+\underline{e})} (\ln(1+\alpha) - \ln(\alpha+\underline{e})) - 2(\ln(1+\alpha) - \ln(\alpha+\underline{e}))^3.$$

Note, first, that β_0 is positive.

$$\beta_0 > 0 \Leftrightarrow \frac{(1 - \underline{e})^2}{(1 + \alpha)(\alpha + \underline{e})} > \ln\left(\frac{1 + \alpha}{\alpha + \underline{e}}\right)^2 \Leftrightarrow \frac{(1 - \underline{e})}{\sqrt{(1 + \alpha)(\alpha + \underline{e})}} > \ln\left(\frac{1 + \alpha}{\alpha + \underline{e}}\right).$$

Denote $t \equiv \frac{1 + \alpha}{\alpha + \underline{e}}$. Then

$$\beta_0 > 0 \Leftrightarrow \frac{t - 1}{\sqrt{t}} > \ln t.$$

Given $\partial(\frac{t-1}{\sqrt{t}} - \ln t)/\partial t = \frac{(\sqrt{t}-1)^2}{2t\sqrt{t}}$, which is strictly positive for $t > 1$ and given $\frac{1-1}{\sqrt{1}} - \ln 1 = 0$, it follows that $\frac{t-1}{\sqrt{t}} > \ln t$ and $\beta_0 > 0$. Therefore, at low levels of precision, marginal increases in transparency improve expected utility.

Next, consider β_1 . Note that

$$\beta_1 = -\frac{(t-1)^3(t+1)}{t^2} + 4\frac{(t-1)^2}{t} \ln t - 2(\ln t)^3$$

Denote $g(t) \equiv t^3 \cdot \beta_1'(t)$. Sign of $g(t)$ is, then, equal to the sign of $\beta_1'(t)$. Note that $g(t)$ is such that $g(1) = g'(1) = g''(1) = g'''(1) = 0$, $g^{(4)}(1) = -20$, and, importantly,

$$g^{(5)}(t) = -24\frac{t^2 - 1 + 2t \ln t}{t^4}.$$

For all $t > 1$, $g^{(5)}(t) < 0$. Because the highest-order non-zero derivative at the boundary is negative and the function continuously decelerates, $g(t) < 0$ for all $t > 1$. Consequently, $\beta_1'(t) < 0$. Since $\beta_1(1) = 0$ and the function is strictly decreasing, $\beta_1(t) < 0$ for all $t > 1$.

Because the comparative static is linear in q , takes a strictly positive intercept ($\beta_0 > 0$), and a strictly negative slope ($\beta_1 < 0$), there exists a unique threshold $\bar{q}$ such that

$$\frac{\partial EU^V}{\partial q} \begin{cases} > 0, & \text{if } q < \bar{q} \\ < 0, & \text{if } q > \bar{q} \end{cases}$$

where $\bar{q} \equiv -\frac{\beta_0}{\delta\beta_1} > 0$.

Proof of Proposition 10

Let $I_k \equiv \mathbb{E}[(\alpha + e^I)^{-k}]$. Differentiating with respect to α yields $\frac{\partial I_k}{\partial \alpha} = -kI_{k+1}$. Applying this to the expected voter utility gives

$$\frac{\partial EU^V}{\partial \alpha} = \frac{1}{16}(3I_4 - 2I_2^2 - 4I_1I_3 + 3I_1^2I_2)\delta^2q^2 - \frac{3}{4}(I_3 - I_1I_2)\delta q + \frac{1 + \delta}{2}I_2.$$

This is parabola. We first establish the signs of its coefficients. $\frac{1+\delta}{2}I_2 > 0$ because $I_2 > 0$. The linear coefficient $\frac{3}{4}(I_3 - I_1I_2) > 0$ because $I_3 - I_1I_2 = \text{Cov}((\alpha + e^I)^{-1}, (\alpha + e^I)^{-2})$ and both arguments are strictly positive and monotonically related. To determine the sign of $\frac{1}{16}(3I_4 - 2I_2^2 - 4I_1I_3 + 3I_1^2I_2)$, let $t \equiv \frac{1+\alpha}{\alpha+\underline{e}} > 1$. The sign of $\frac{1}{16}(3I_4 - 2I_2^2 - 4I_1I_3 + 3I_1^2I_2)$ depends strictly on

$$f(t) \equiv (t^2 - t + 1)(t - 1)^2 - 2t(t^2 - 1) \ln t + 3t^2(\ln t)^2$$

Evaluating $f(t)$ and its derivatives at the boundary yields $f(1) = f'(1) = f''(1) = f'''(1) = 0$ and $f^{(4)}(1) = 10$. Because the fifth derivative, $f^{(5)}(t) = \frac{12}{t^4}(t^2 - 1 + 2t \ln t)$, is strictly positive for all $t > 1$, it follows that $f(t) > 0$. Therefore, $\frac{1}{16}(3I_4 - 2I_2^2 - 4I_1I_3 + 3I_1^2I_2) > 0$.

Thus, $\frac{\partial EU^V}{\partial \alpha}$ is a strictly convex quadratic in q . Because $e^I \geq \underline{e}$, we can bound $I_3 < (\alpha + \underline{e})^{-1}I_2$. Therefore,

$$\frac{\partial EU^V}{\partial \alpha} > I_2 \left(\frac{1 + \delta}{2} - \frac{3\delta q}{4(\alpha + \underline{e})} \right).$$

As $\alpha \rightarrow \infty$, the term $\frac{3\delta q}{4(\alpha + \underline{e})} \rightarrow 0$. Therefore, $\exists \bar{\alpha}$ such that for all $\alpha > \bar{\alpha}$, $\frac{\partial EU^V}{\partial \alpha} > 0$. Consider

$$\min_q \frac{\partial EU^V}{\partial \alpha} = \frac{1 + \delta}{2} I_2 - \frac{(\frac{3}{4}(I_3 - I_1I_2)\delta)^2}{4\frac{1}{16}(3I_4 - 2I_2^2 - 4I_1I_3 + 3I_1^2I_2)\delta^2} = \frac{1 + \delta}{2} I_2 [1 - \Phi(\alpha)],$$

where

$$\Phi(\alpha) \equiv \frac{9(I_3 - I_1I_2)^2}{2(1 + \delta)I_2(3I_4 - 2I_2^2 - 4I_1I_3 + 3I_1^2I_2)}.$$

Note that $\Phi(\alpha) \propto \alpha^{-2}$ and $\lim_{\alpha \rightarrow \infty} \Phi(\alpha) = 0$ (thus, $\frac{\partial EU^V}{\partial \alpha}$ always positive at $\alpha \rightarrow \infty$) and $\Phi(\alpha)$ is strictly monotonically decreasing in α . It remains to show that

$$\lim_{\alpha \rightarrow 0, \delta \rightarrow 0, \underline{e} \rightarrow 0} \Phi(\alpha) = 9/8 > 1.$$

Because $\Phi(\alpha)$ is strictly monotonically decreasing from $\Phi(0) > 1$ to $\Phi(\infty) = 0$, there exists a unique, globally defined threshold $\hat{\alpha}$ such that $\Phi(\hat{\alpha}) = 1$. For all $\alpha < \hat{\alpha}$, $\frac{\partial EU^V}{\partial \alpha} < 0$.

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